

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	micro	radio
Energy (E)	$\uparrow$						$\downarrow$
Frequency (ν)	$\uparrow$						$\downarrow$
Wavelength (λ)	$\downarrow$						$\uparrow$
Wavelength (λ)	<0.01	0.01-10	10-400	400-700	700-1e6	1e6-9	>1e9

• violet • indigo • blue • green • yellow • orange • red

(2) $c = \lambda\nu$ (Speed of light = wavelength $\times$ frequency)

c	ν (Hz)	kHz	MHz	GHz
λ (m)	2.9979e8 (m/s)	2.9979e5 (m·kHz)	2.9979e2 (m·MHz)	2.9979e-1 (m·GHz)
nm	2.9979e17 (nm·Hz)	2.9979e14 (nm·kHz)	2.9979e11 (nm·MHz)	2.9979e8 (nm·GHz)

(3) $E = h\nu = hc/\lambda$ (Energy = Planck $\times$ frequency)

hc , h	λ (m)	λ (nm)	ν (Hz)
E (J / photon)	1.9864e-25 (J·m)	1.9864e-16 (J·nm)	6.626e-34 (J·s)
J / mol	1.1962e-1 (J·m/mol)	1.1962e8 (J·nm/mol)	3.9902e-10 (J·s/mol)
kJ / mol	1.1962e-4 (kJ·m/mol)	1.1962e5 (kJ·nm/mol)	3.9902e-13 (kJ·s/mol)

• Grams $\rightarrow$ mol $\rightarrow \times 6.022e23$ for photon count

(4) Quantum Principles

Maxwell	Light is electromagnetic radiation
Planck	Energy emitted only in fixed amounts (quanta) (black-body radiation)
Einstein	Different light has different amounts of fixed energy (photons) (photoelectric effect)
Bohr	Electrons are quantum (line spectra); Rydberg eq.
De Broglie	Electrons are particle and wave (5)
Schrodinger	Wave equation (energy and spatial prob. of e-) (n, l, m_l, m_s)
Heisenberg	Position and velocity not both knowable

• $E = h\nu = hc/\lambda = R(1/n_f^2 - 1/n_i^2)$

• $n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}$; $n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) $\lambda = h/(m \cdot v)$ (De Broglie)

- 1 nm = 1e-9 m
- 1 e- = 9.11e-31 kg

h	u (m/s)	km/s
m (kg)	6.626e-34 (J·s)	6.626e-37 (kJ·s)
g	6.626e-31 (mJ·s)	6.626e-34 (J·s)
amu	3.9892e-7 (J·s·amu/kg)	3.9892e-10 (kJ·s·amu/kg)

(6) Electron Configuration (Aufbau, Pauli, Hund)

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p
- Round-robin sublevel before double occupaton (Hund)
- Two e- per orbital (Pauli's exclusion)

Symbol	Name	Value
n	Principal energy level (shell)	(1, 2, 3, ...)
l	Sublevel = subshell (shape)	(0, ..., n - 1)
m_l	Orbital (orientations)	(-l, ..., l)
m_s	Spin (2 per orbital)	$\pm 1/2$

Sublevel	l	m_l	# orbitals	# e-
s	0	0	1	2
p	1	± 1	3	6
d	2	± 2	5	10
f	3	± 3	7	14

Element	EC	Element	EC
Cr	[Ar]4s ¹ 3d ⁵	Ag	[Kr]5s ¹ 4d ¹⁰
Mo	[Kr]5s ¹ 4d ⁵	Au	[Xe]6s ¹ 4f ¹⁴ 5d ¹⁰
Cu	[Ar]4s ¹ 3d ¹⁰	Pd	[Kr]4d ¹⁰ (no 5s)

(7)-(8) Reading EC, Anion, Cation

EC $\rightarrow$ element	add every superscript to get Z
Paramagnetic	at least 1 unpaired val. e- (else dia.)
Ground state	follows Aufbau (else excited)
Valence electrons	highest / outer shell (else core)
Anion	count val. e- needed until noble gas
Cation	remove val. e- until noble gas
Tx metal	remove s before d
Isoelectric	same EC (e.g. Na ⁺ , Al ³⁺ , F ⁻ , N ³⁻ , Ne)

(9) Quantum Numbers, Penetration & Shielding

- Shell n holds n^2 orbitals, $2n^2$ e-
- **energy: $s < p < d < f$** (within shell, same n)
- **Pauli exclusion** — no two e- carry the same four quantum numbers $\rightarrow$ 2 per orbital, opposite spins
- **Penetration** — orbital reaching into the core region
- $\uparrow$ shielding $\rightleftharpoons \uparrow$ reduced $Z_{eff} \rightleftharpoons \downarrow$ penetration $\rightleftharpoons \uparrow$ **energy** (reverse also holds)
- **Shielding** — only **core** (inner) e- shield effectively; valence e- shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e- = total e- - valence e- (= the **previous noble gas**)
2. $Z_{eff} \approx Z - \text{core } e^-$
 - Same period $\rightarrow$ same core, so the **rightmost** element wins
 - Poor valence shielding is why Z_{eff} rises across a period

(11)-(13) Trends

	Trend	Top	Bot	Left	Right
(11) Atomic size (radius)	$\swarrow$	$\downarrow$	$\uparrow$	$\uparrow$	$\downarrow$
(12) Ionization energy	$\nearrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\uparrow$
(13) Electron affinity	$\nearrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\uparrow$
Z_{eff}	$\rightarrow$	$\sim$	$\sim$	$\downarrow$	$\uparrow$
Metallic	$\swarrow$	$\downarrow$	$\uparrow$	$\uparrow$	$\downarrow$
Nonmetallic	$\nearrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\uparrow$
Electronegativity (EN)	$\nearrow$	$\uparrow$	$\downarrow$	$\downarrow$	$\uparrow$

Atomic size

- Down a group **bigger** (new shell) • across a period **smaller** (Z_{eff} pulls in)
- Group position beats period position when both apply
- cation < neutral < anion

Ionization energy

- He $\uparrow$ of all • opposite of radius
- $IE_2 > IE_1$ always
- **Big jump** appears once the first **core** e- is removed

Electron affinity

- **Most eager for an e- = halogens, upper right** — this course writes it both as "most negative EA " and "largest positive EA "; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- **$EA(X)$ and IE of X are the same magnitude** — F + e- releases exactly what F $\rightarrow$ F costs

(14) Families & Diagonal Rule

Family	Name	Reactivity	e-	Charge
1A	alkali metals	most reactive metals (downward)	lose 1	1+
2A	alkaline earth	reactive, less than 1A	lose 2	2+
7A	halogens	reactive nonmetals (upward)	gain 1	1-
8A	noble gases	inert	—	0

- **Lanthanide** (58–71), **Actinide** (90–103)
- **Diagonal rule** (e.g. Li~Mg, Be~Al, B~Si)
- **Diatomic** (H₂, N₂, O₂, F₂, Cl₂, Br₂ (l), I₂ (s))

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K₂O, MgO, Fe₂O₃, Cr₂O₃) · nonmetal oxide = **acidic** (P₄O₁₀, SO₂, CO₂, Cl₂O₇) · Al₂O₃ and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3- · 6A 2- · 7A 1-; criss-cross and reduce

• Same-group swap keeps the pattern: BaS → **CaO**

• 2A + 3F₂ → 2AF₃ ⇒ A is 3+ (**Al**) · 3Li + Z → Li₃Z ⇒ Z is 3-, so Mg gives **Mg₃Z₂**

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH₄NO₃, K₂SO₄); all-nonmetal = covalent only

Family reactions

- 2K(s) + 2H₂O(l) → 2KOH(aq) + H₂(g)
- H₂(g) + Cl₂(g) → 2HCl(g)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first

2. If charges tie, smaller ions win

- ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ **U**

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read *EN* off the chart — *EN* ↑ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)
2. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar"** → ΔEN **alone** (largest difference wins)
- **"Strongest / shortest bond"** → **bond order FIRST**: triple > double > single. ΔEN only breaks ties between equal orders — a C≡N beats a high- ΔEN Si=O
- Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. *FC* on every atom
 2. Check ΣFC = the overall charge
 3. Pick the structure with *FC*'s **nearest 0**
 4. Put any -1 on the **more electronegative** atom
- Equivalent resonance forms → real bonds are the **average** (e.g. 1½)

(19) Lewis Structures x4

1. Total valence e- — +1 per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e- onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e- per atom (H wants 2) — H ₂ O, CH ₄ , NH ₃ , CO ₂ , CH ₂ Cl ₂ , CH ₃ CH ₂ Cl
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. BF ₃ /BCl ₃ /AlI ₃ leave 6 e- on the center; BeCl ₂ leaves 4
(c) Expanded octet	Central atom period 3 or below (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — AsCl ₅ , XeI ₂ , SO ₄ ²⁻
(d) Organic	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH₂ amine

- Every C makes 4 bonds; a -COOH carbon carries one =O and one -O-H; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero *FC*, add double bonds to zero it (e.g. two S=O in SO₄²⁻)

(20) Photoelectric Effect (*hν* - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

Variable	Value
<i>h</i>	6.626e-34 J·s
<i>m_e</i>	9.11e-31 kg
<i>hc</i>	1.9864e-25 J·m
2 <i>hc/m_e</i>	4.3609e5 m ³ /s ²

Have	Want	Use
λ, λ_{thr}	<i>u</i>	$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
ν_{thr} or λ_{thr}	Φ, Φ / mol	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$, n.b. $\Phi \times N_A$ (/1e3 for kJ/mol)
Φ, λ	<i>u</i>	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)
 2. $\Delta x \geq h/(4\pi \cdot \Delta p)$
- Result is the **minimum** uncertainty
 - Rearrange the same way to solve for Δu
 - Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), *IE* (+), bond dissociation (+), *EA* (usually -), *U* (-)
 2. **Halve** the bond dissociation if you need only one atom
 3. Chain: elements → gaseous atoms → gaseous ions → solid
 4. Set the chain equal to ΔH°_f
 5. Solve for the single unknown
- Use *IE₂* as well when the cation is 2+
 - U runs ions(g) → solid, so it comes out **negative**; report |U| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
 2. $\Sigma BE(\text{broken})$, always positive
 3. $\Sigma BE(\text{formed})$, subtracted
 4. $\Delta H = \Sigma \text{broken} - \Sigma \text{formed}$
- Bonds unchanged on both sides cancel — skip them
 - Negative ΔH = exothermic
 - For "heat released by *g* of X," convert *g* → mol and scale